Sabrina Weng

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Education

M.S. Bioengineering — University of Pennsylvania Philadelphia, PA
Concentration in Data Science and Machine Learning *December 2024*
Coursework: ML, AI, Big Data, Biomedical Image Analysis, Medical Device Management and Strategy (Wharton).

B.S. Biomedical Science and Engineering — National Central University Taoyuan, Taiwan
Specialize in Biomedical Image Processing and Machine Learning Modeling *June 2023*
Publication: IEEE Transactions on Emerging Topics in Computational Intelligence, 2024

Skills

Programming: Python (NumPy, Pandas, scikit-learn, TensorFlow, PyTorch, OpenCV), SQL, Git/GitHub, Cloud (AWS)
Data Science & Analytics: Exploratory Data Analysis (EDA), Data Cleaning and Preprocessing, Feature Engineering, Data Integration, ETL/ELT, Hypothesis Testing, Statistical Analysis, Data Visualization (Power BI, Tableau, Matplotlib)
Machine Learning & AI: Predictive Modeling, Neural Networks, Hyperparameter Tuning, Model Evaluation
Soft Skills: Leadership, Problem-Solving, Teamwork, Attention to Detail, Creativity, Data-driven Decision Making, Project Management, Accountability, Adaptability, Influencing and Communication

Experience

Data Scientist Intern — Children's Hospital of Philadelphia Philadelphia, PA
Deep Learning Model Development and Data Analysis *May 2024 – August 2024*
- Developed scalable ML pipelines for large-scale multimodal datasets, including multiparametric MRI scans and structured clinical data, enabling faster processing and scalable experimentation.
- Ran experiments comparing baseline models with attention-augmented deep learning, improving prediction by 15%.
- Translated complex model outputs into risk stratification insights that informed decision-making for clinicians by highlighting patterns linked to treatment planning.

Product Marketing — Penn Health-Tech Philadelphia, PA
Product Strategy and Commercialization *February 2024 – May 2024*
- Led planning and execution of product strategy, coordinating cross-functional teams and integrating data-driven insights to advance a product from concept to market.
- Conducted data-driven market analysis, competitive benchmarking, and evaluation of regulatory/reimbursement pathways to inform decision-making, strategy and reduce commercialization risk.
- Presented to senior stakeholders, demonstrating effective communication of complex results into actionable insights.

ML Research Assistant — NCU Geometric Data Vision Laboratory Taiwan
Model Optimization and Medical Image Analysis *September 2021 – May 2023*
- Designed and implemented optimized neural network architectures for large-scale data, reducing computation time and achieving 92%+ accuracy on benchmark datasets.
- Built preprocessing techniques that improved accuracy to 99% and ensured reproducibility.
- Directed a team of five in an international competition, leading execution to secure 1st place out of 93 teams.

Projects

Data Analyst
Machine Learning Model Development for ICU Mortality Prediction *March 2024 – May 2024*
- Built end-to-end predictive workflow on structured ICU data with EDA, feature engineering, and model training.
- Boosted model accuracy by 10% using Neural Networks with domain-specific metrics, ensuring model reliability.

Data Scientist
Exploratory Data Analysis and Predictive Modeling of Protein Atom Ice-Philicity *September 2024 – December 2024*
- Performed EDA with hypothesis testing, anomaly detection, and feature engineering to identify key predictors.
- Trained models with class imbalance handling, enhancing accuracy by 27% over the baseline model.